ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

Project Proposal Report

SWE 4304 : Software Project Lab I

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Project Title: Dormatrix

Introduction To Our Project:

Dormatrix is a Command Line Application featured on digitalizing IUT Dormitory System. This application addresses the problems faced in IUT Female Dormitory and provides a centralized solution. To better understand how Dormatrix transforms hall life, let us step into the experience of Tayma, a resident of the IUT Female Hall.

Motivation behind the Project:

There are a few problems that residents face as they reside in the IUT Female Hall:

- **Difficulty in submitting complaints:** The process of submitting and resolving complaints is long and involves many steps, and there is never a clear update provided on their status.

- **No organized facility booking:** In the study room, students tend to keep their bags on seats to "reserve" them without actually using the space, leaving no available place for residents to sit.

- **No proper system for the departmental store:** Some students buy items on credit from the departmental store and later forget to pay, creating an unmanaged system. Also the students find it difficult to access the available inventories.

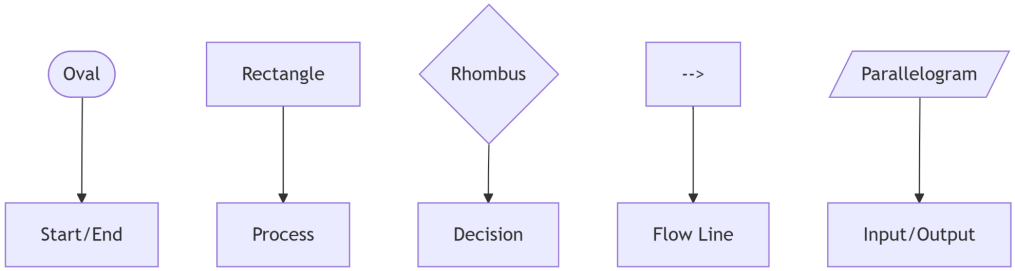
- **Shared appliances are not managed well:** Finding space in the refrigerator becomes difficult for them since proper usage protocols are not enforced among the students.

- **Lack of food management infrastructure:** The yearly payment system remains in place instead of a token-based method, even though residents often do not eat hall food.

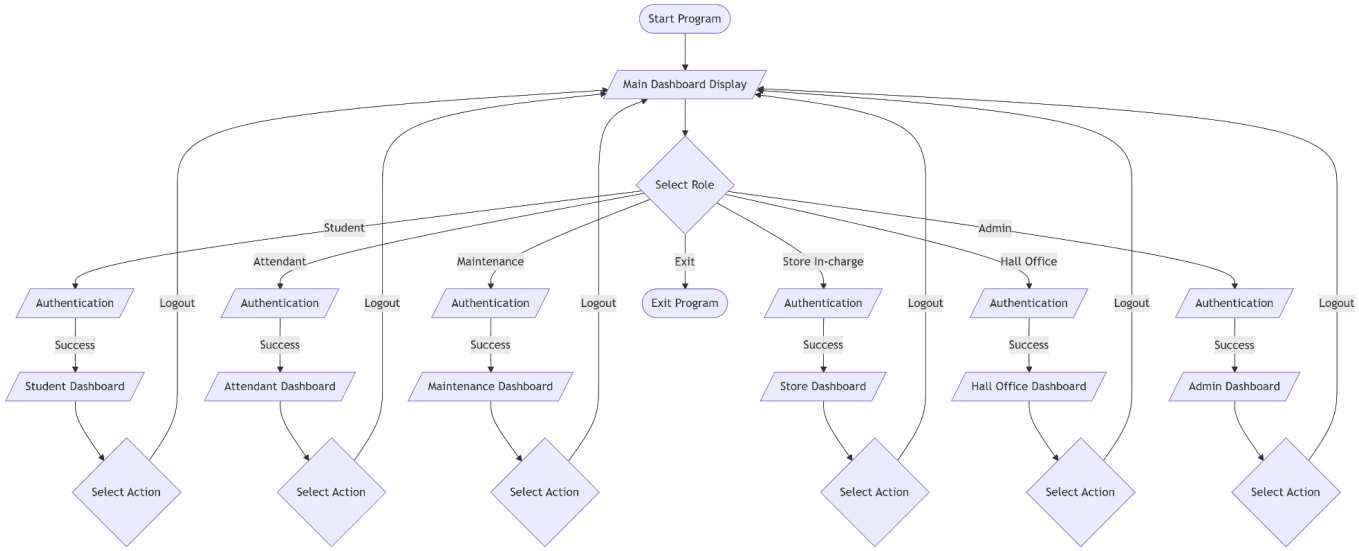
Hence, we came up with a solution that would benefit them.

Flow Chart

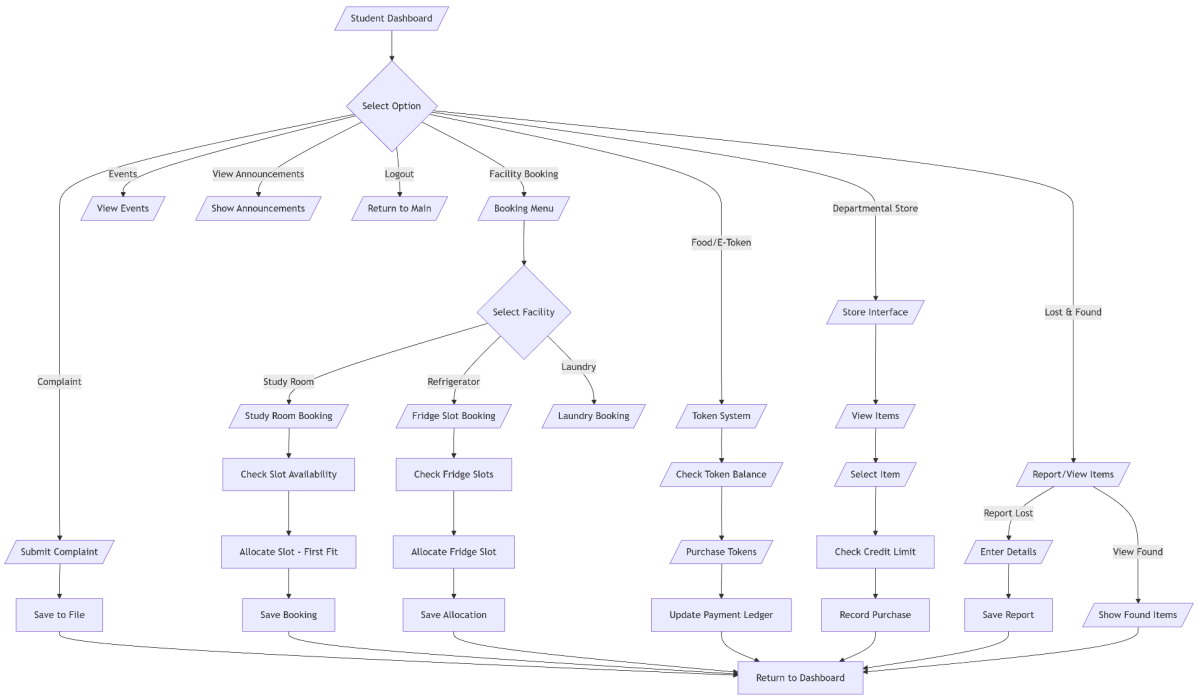
Flowchart Symbols Legend:



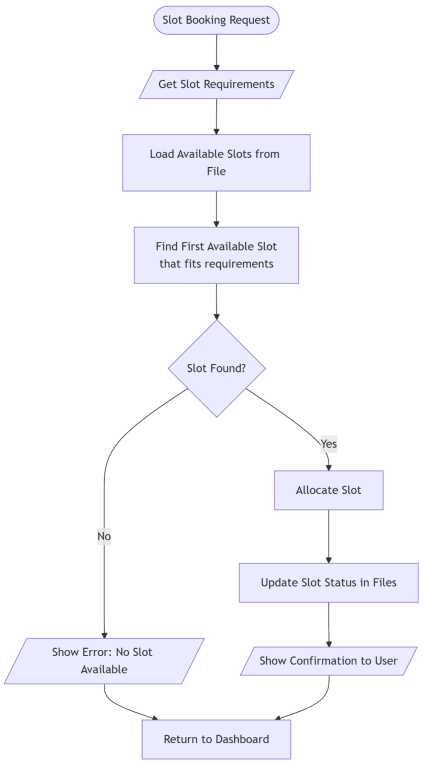
1. **System Overview:**



2. Student Flow Chart Details:



3. Slot Allocation Algorithm Flow



4. Complain Management Flow:

Key Features And User Walk Through:

Residents of IUT female hall can begin their day by logging into Dormatrix, the new command-line operating system designed for hall management.

For Residents :

Student Dashboard – Personalized interface showing all hall services

- ➔ Room Dashboard
- ➔ Complaint Submission
- ➔ Facility Booking (Study Room Slot, Refrigerator Slot)
- ➔ Food
- ➔ Departmental Store
- ➔ Lost & Found
- ➔ Events
- ➔ Meal E-Token Purchase

For Attendants

- ➔ Centralized hall overview
- ➔ Easy complaint monitoring
- ➔ Bulk announcement system
- ➔ Facility booking oversight

For Maintenance Staff

- ➔ Priority-based task queue
- ➔ Clear complaint lists
- ➔ Status update ability
- ➔ Remaining task view

For Store In-Charge

- ➔ Account purchase & payment records
- ➔ Inventory tracking
- ➔ Sales summaries

For Hall Office

- ➔ Resident information management
- ➔ Easy report generation

For System Administrators

- ➔ User account management
- ➔ Permission controls
- ➔ Backup and logging

Technical Mechanisms

Password Authentication

- ❖ Secure login system
- ❖ Role-based access control

Slot Memory Management

- ❖ Slots as memory frames
- ❖ First-fit algorithms

This is be implemented in:

- ➔ Fridge slots allocation
- ➔ Study room seat booking
- ➔ Laundry machine scheduling

Permission Mapping

- ❖ 6 user roles with distinct permissions
- ❖ Role: Student, Attendant, Maintenance, Store, Office, Admin
- ❖ Permission inheritance and override

Due Ledgers System

- Payment tracking
- Inventory tracking

Service Schedule Tracker

- Staff assignment optimization
- Confirms task completion

Tools and Technologies

- **Language :** Java
- **File Based Storage :** text based file
- **I/O:** Java FileReader / FileWriter / BufferedReader
- **Architecture:** Modular OOP
- **Version Control:** Github
- **IDE:** IntelliJ, VS Code

Proposed Timeline

Week 4 — Dashboard Foundation

- Build CLI skeleton, menus, navigation
- Role selection
- Role Based Dashboards

Week 5 —6 Authentication & User System and Room Dashboard & Complaint System

- User model
- Login & logout
- Role-based redirection to dashboards
- Load room info
- Student complaint submission
- Attendant: view complaints

- Maintenance: priority handling

Week 7 —8 Facility Booking System and Laundry & Permission Mapping

- Study room seat booking
- Fridge slots with first-fit allocation
- Laundry machine slot scheduler
- Define permissions for 6 roles, implement inheritance/override

Week 9 —10 Departmental Store Module and Food & Meal E-Token System

- Inventory management & sale tracking
- Track dues & generate sales summaries
- Student purchase + view/pay dues
- Buy & validate meal tokens

Week 11 — Lost & Found, Events

- Add lost/found items
- Announcements & hall events

Week 12 — Backend Enhancement

- Optimize file handling & parsers
- Backup & logging system
- Error/exception handling
- Organize controllers/models

Week 13 — Testing & Debugging

- Unit Testing

Week 14 — Documentation & Final Presentation

- Ensure Proper Documentation of Code
- Video Demo
- Final Presentation

Additional Planning

Estimated Library Functions

- **java.io (File Handling) :** reader and writer
- **java.util.Scanner:** system.in and system.out

Note: We are planning to manually implement various libraries like List, Hashing etc.

Folder Structure and Packages

- ➔ libraries : Having all the custom Libraries we plan to implement
- ➔ models
- ➔ exceptionHandlings
- ➔ controllers (backend + database managers)
- ➔ cli
- ➔ data (all text files)
- ➔ main

SDLC Model For the Project

Iterative + Incremental Model

Iterative :

Building a feature then adding functionalities in that. The functionalities are implemented iteratively.

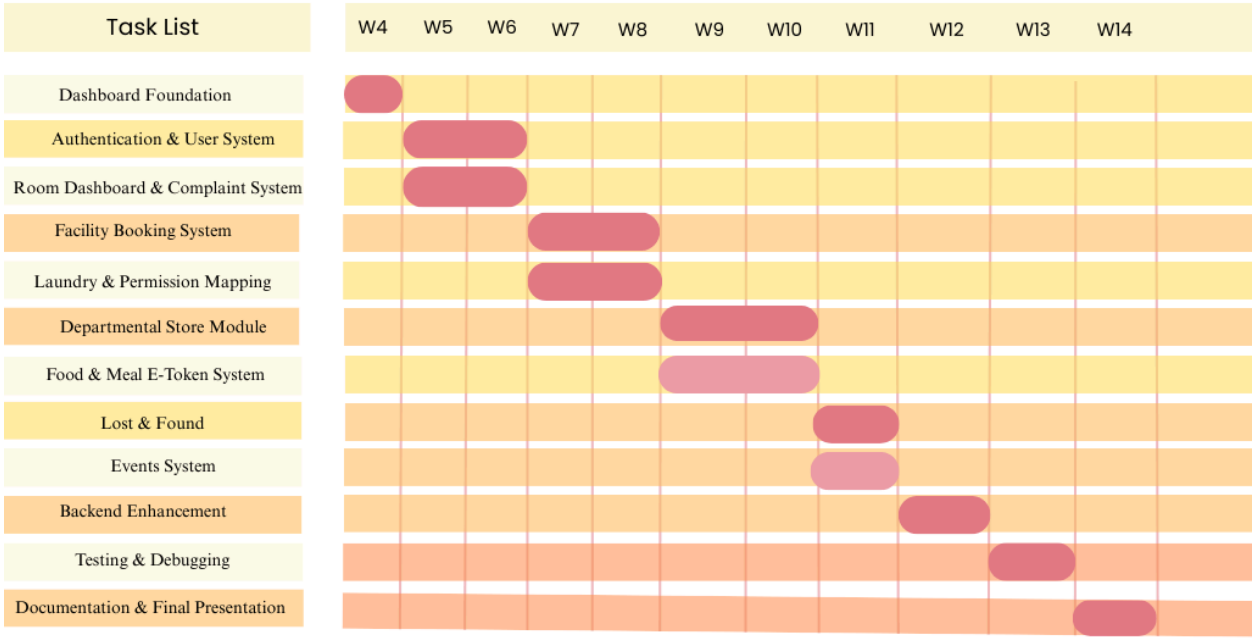
Incremental :

Building module by module or feature by feature

Action taken based on the feedback:

In response to the feedback, we have provided an updated Gantt chart that clarifies the timelines and outlines the distribution of work.

Gantt Chart on the Proposed Timeline:



Conclusion:

In summary, Dormatrix delivers a structured and efficient solution for managing essential dormitory services.